

RODRIGO GAYTAN MIRANDA

Mechatronics Engineering | Industrial Automation, Electrical Systems & Industrial Operations

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SUMMARY

Recently graduated Mechatronics Engineer at Tecnológico de Monterrey with technical commercial experience gained at ABB México, supporting product specification, quotations and customer-facing sales processes for industrial motor solutions. Developed 10+ bilingual technical sales manuals for NEMA/IEC/Baldor motor lines, reducing response times by 10% and directly supporting commercial teams. Combines engineering fundamentals with a PMI-based project management specialization (Rice University) and international business exposure through an academic exchange in Budapest.

WORK EXPERIENCE

Trainee specialist - Low and Medium Voltage Motors | ABB México

Jan 2025 - Jul 2025

- Prepared technical quotations and supported product selection for low and medium voltage motor solutions, acting as a technical point of contact between engineering and commercial teams.
- Authored 10+ bilingual technical sales manuals for NEMA, IEC and Baldor motor lines, cutting technical response times by 10% and equipping the sales team to answer customer questions faster.
- Supported customer facing quotation and specification processes, translating technical motor specifications into commercially actionable information for sales representatives.

ACADEMIC PROJECTS

Structures Senior - SAE Aerodesign | Tecnológico de Monterrey

Aug 2024 - Jun 2026

- Contributed to improving team performance from 22nd to 18th place among 33 teams at SAE Aerodesign East 2025 through structural design and prototype development.
- Designed and performed structural analysis of winglets to improve aerodynamic efficiency and flight performance of a remote controlled aircraft prototype.
- Remote/advisory role during academic exchange Aug 2025 Jan 2026.

Flexible Single-link Robot Control System | CINVESTAV

Jan 2026 - Jun 2026

- Built and programmed a robot, implementing a Python-based RL agent for a flexible beam mathematical model to autonomously tune optimal PID controller gains.
- Achieved a 11% efficiency improvement in system response, successfully suppressing structural vibrations and reducing settling time.
- Developed custom Python scripts to process, analyze, and visualize real-time training metrics and control system performance.

PLC Programming- Scholarship Service | Tecnológico de Monterrey

Aug 2024 - Dec 2024

- Programmed and implemented industrial automation systems using Siemens S7 PLCs through TIA Portal, developing control logic with Function Block Diagram (FBD) programming.
- Integrated industrial communication networks using Profinet while designing automation systems involving pneumatic actuators, electrical actuators, HMIs and manufacturing cells.
- Developed automation and motion control applications including PID-controlled systems and motor control through variable frequency drives (VFDs), validating system integration and operational performance.

EDUCATION

B.S. in Mechatronics Engineering | Tecnológico de Monterrey - State of México Campus

Aug 2022 - Jun 2026

- Relevant coursework: Industrial automation, control systems, robotics, machine design, electronics.
- EGS Leadership Program: selective program focused on human leadership, entrepreneur spirit, and institutional values
- GPA : 88/100

International Exchange Program | Budapest University of Technology and Economics, Hungary

Aug 2025 - Jan 2026

- Coursework: Management & Business economics, Quality Management, electrical machines, renewable energy and manufacturing processes.

TECHNICAL SKILLS

- **Automation & control:** PLC - Siemens TIA Portal (FBD,LAD), Emulate3D
- **Engineering Design:** SolidWorks (intermediate), CATIA (intermediate), AutoCad (intermediate), Proteus 8
- **Simulation & Analysis:** Python (OpenCV, data analysis), C++ (basic), Power BI, Office 365, Simulink, MATLAB, Excel (advanced), Jira
- **Motor technologies:** NEMA,IEC low and medium voltage, technical specification and quotation
- **Methodologies:** Industry 4.0, project management (PMI framework), quality management systems
- **Languages:** Spanish (native), English (B2+)

CERTIFICATIONS

- Introduction to Cybersecurity | CISCO - 2025
- Engineering Project Management Specialization | Rice University (2026)